



Project Title: -

Online Shopping Management System

Course: -

Database system

Submitted By: -

Group E

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Online Shopping Management System

Project Description

The Online Shopping Management System is a database-based software project developed to manage online shopping activities efficiently. This system will store and organize information related to customers, products, categories, orders, payments, and deliveries. The purpose of this project is to provide a digital platform for managing an online store in a fast, secure, and organized way instead of using manual records.

What We Are Going to Build

We are going to build an Online Shopping Management System that allows customers to browse products, place orders, and make online payments. The system will also help administrators manage products, customer records, order details, payment information, and delivery status through a centralized database system.

The system database will include entities such as:

- Customer
- Product
- Category
- Order
- OrderItem
- Payment
- Delivery

Main Features

- Customer registration and login system
- Product and category management
- Shopping cart functionality
- Online order placement
- Secure payment management
- Delivery and order tracking
- Product search and filtering
- Stock and inventory management
- Database record maintenance and updates

Expected Outcome

After completing this project:

- Online shopping operations will become more efficient and organized
- Product, customer, and order records will be stored securely
- Manual paperwork and data redundancy will be reduced
- Customers will get a convenient shopping experience
- Administrators will easily manage products, orders, and payments
- The system will improve accuracy, speed, and data management

This project will also enhance practical knowledge of database management systems, ER diagrams, relational schemas, normalization, and SQL operations.

Project Scenario

The Online Shopping Management System is designed to manage and automate the activities of an online store. The system focuses on storing, organizing, and managing data related to customers, products, categories, orders, payments, and deliveries through a centralized database system.

The project scenario involves creating a platform where customers can browse products, add items to a shopping cart, place orders, and make payments online. At the same time, administrators will be able to manage products, monitor stock levels, track customer orders, handle payments, and manage delivery information efficiently.

Requirements

The main requirement of the system is to provide a secure, accurate, and user-friendly environment for online shopping management. The database must support:

- Customer registration and login
- Product and category management
- Order processing
- Payment management
- Delivery tracking
- Inventory management
- Data storage and retrieval

The system will use entities such as Customer, Product, Category, Order, OrderItem, Payment, and Delivery with proper relationships and constraints to ensure data consistency and reduce redundancy.

The expected result of this project is a fully organized database system that improves efficiency, reduces manual work, and provides better management of online shopping operations.

Entities, Attributes, and Relationships

1. Customer

Attributes

- CustomerID (PK)
- FirstName
- LastName
- Email
- Password
- Phone
- Address
- RegistrationDate

2. Product

Attributes

- ProductID (PK)
- ProductName
- Description
- Price
- StockQuantity
- Brand
- CategoryID (FK)

3. Category

Attributes

- CategoryID (PK)
- CategoryName
- CategoryDescription

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4. Order

Attributes

- OrderID (PK)
- CustomerID (FK)
- OrderDate
- TotalAmount
- OrderStatus

5. OrderItem

Attributes

- OrderItemID (PK)
- OrderID (FK)
- ProductID (FK)
- Quantity
- SubTotal

6. Payment

Attributes

- PaymentID (PK)
- OrderID (FK)
- PaymentDate
- PaymentMethod
- PaymentStatus
- Amount

7. Delivery

Attributes

- DeliveryID (PK)
- OrderID (FK)

- DeliveryAddress
- DeliveryDate
- DeliveryStatus

Relationships

Relationship	Type
Customer places Order	One-to-Many
Category contains Product	One-to-Many
Order contains OrderItem	One-to-Many
Product appears in OrderItem	One-to-Many
Order has Payment	One-to-One
Order has Delivery	One-to-One

Many-to-Many Relationship

Order ↔ Product

- One order can contain many products
- One product can appear in many orders
- This relationship is resolved using the OrderItem entity.

Order >—< Product

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OrderItem

Primary Keys (PK)

- CustomerID
- ProductID
- CategoryID
- OrderID
- OrderItemID

- PaymentID
- DeliveryID

Foreign Keys (FK)

- CategoryID in Product
- CustomerID in Order
- OrderID in OrderItem
- ProductID in OrderItem
- OrderID in Payment
- OrderID in Delivery

Entity Relationship Diagram



